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System and method of optical antenna tuning

Abstract

A method includes determining an operating frequency of an antenna at a component of the antenna. The method includes determining, at the component of the antenna, a filter node of a plurality of filter nodes to activate to tune the antenna to the operating frequency. The method includes transmitting power to the filter node, wherein the power is transmitted via a first optical fiber. The method also includes sending a signal from the component to a light source. The activation of the light source sends an optical signal to the filter node. The filter node adjusts a characteristic of a radiating element coupled to the filter node using the power responsive to the optical signal. Adjustment of the characteristic facilitates tuning the antenna to the operating frequency.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/499,442, filed on Oct. 12, 2021, which is a divisional of U.S. patent application Ser. No. 15/977,169, filed on May 11, 2018 (now U.S. Pat. No. 11,177,574), which is a continuation of U.S. patent application Ser. No. 14/084,200, filed Nov. 19, 2013 (now U.S. Pat. No. 10,003,131), which are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure is generally related to optical antenna tuning.

BACKGROUND

(2) Antennas are used as coupling elements between circuits (e.g., radio frequency (RF) circuits) and a wireless medium. Antenna design usually uses metallic conductors to transfer energy between electromagnetic waves and electrical currents. An antenna's size is usually related to a target wavelength. For example, an antenna may have a characteristic dimension (e.g., length) that is $\frac{1}{2}$ or $\frac{1}{4}$ of a target wavelength.

(3) Some antennas may be used to operate over a wide operating frequency band. Fixed tuning mechanisms, such as frequency-selective lumped filters (called “traps”), may be used to create multi-band antennas that selectively add reactance to portions of a longer antenna to allow the antenna to operate in frequency bands with shorter wavelengths. Reactance may be added to the portions of the antenna by joining segments of a metallic radiating structure of the antenna using traps or filters. Multi-band antennas may also be created using distributed resonant (LC) element structures.

(4) When a spectral coverage of an antenna is very large, it may be useful to have an antenna that is “active” in the sense that metallic element segments of the antenna may be adjusted remotely. One approach has been to include an extendable telescoping mast in the antenna. A length of the telescoping mast may be varied by using a motor drive system. Using an extendable telescoping mast for an antenna in the microwave region may be impractical, as the antenna may be small and the adjustments may distort the behavior of the antenna.

(5) Another approach to adjust an antenna such that a radiating length is matched to a frequency of interest is to use electrical switches or relays. Control and power circuits used to provide adjustment may disrupt the radiative behavior of the antenna due to parasitic conductors that behave as unwanted radiating elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of a particular illustrative embodiment of an antenna that may be optically tuned;
- (2) FIG. 2 is a diagram of an illustrative embodiment of an active filter node that may be included in the antenna of FIG. 1;
- (3) FIG. 3 is a flowchart to illustrate a particular embodiment of a method of optically tuning the antenna of FIG. 1; and
- (4) FIG. 4 is a block diagram of an illustrative embodiment of a general computer system operable to support embodiments of computer-implemented methods, computer program products, and system components as illustrated in FIGS. 1-3.

DETAILED DESCRIPTION

- (5) Systems and methods of optically tuning an antenna are disclosed. Optical signals (e.g., photonic signals) may be used to actively tune the antenna using an electronic switching device (such as a photosensitive field effect transistor). Metallic radiating segments that form the antenna may be selectively activated or deactivated to tune the antenna to a target frequency of operation. The optical signals may be supplied by a light emitting diode (LED), a laser, or another source. The optical signals may be conveyed to an electronic switching device via a fiber optic cable, a light-guide, or via free-space. The optical signal source may be separated from the antenna by a sufficiently long distance to prevent a field of the antenna from being affected by the optical signal source.
- (6) In a particular embodiment, distributed amplifiers may operate as adjustable filters. The distributed amplifiers may adjust parameters (e.g., inductance, capacitance, and/or gain) of the antenna segments. Power-over-fiber (POF) techniques may be used to power the amplifiers. Additionally, optical fibers may be used to provide control signals to the amplifiers. Moreover, the disclosed techniques may be extensible to mass/miniaturized manufacturing techniques.
- (7) In a particular embodiment, a method includes determining a configuration of an antenna. The method also includes selecting a first light source of a plurality of light sources based on the configuration. The method further includes activating the first light source and transmitting an optical signal to a first active filter node of a plurality of active filter nodes of the antenna. In response to receiving the optical signal, the first active filter node deactivates a corresponding radiating element of a plurality of radiating elements.
- (8) In another particular embodiment, an antenna includes a plurality of active filter nodes and a plurality of radiating elements. Each of the plurality of radiating elements is coupled to a particular active filter node of the plurality of active filter nodes. The antenna also includes a plurality of light sources and a controller. The controller activates a first light source of the plurality of light sources. The first light source, in response to being activated, transmits an optical signal to a first active filter node of the plurality of active filter nodes. The first active filter node deactivates a corresponding radiating element of the plurality of radiating elements in response to receiving the optical signal.
- (9) In another particular embodiment, a computer-readable storage device stores instructions that, when executed by a processor, cause the processor to perform operations including determining a configuration of an antenna based on a particular frequency, selecting a first light source of a plurality of light sources based on the configuration, and activating the first light source. The first light source, in response to being activated, transmits an optical signal to a first active filter node of a plurality of active filter nodes of the antenna. The first active filter node deactivates (or adjusts) a corresponding radiating element of a plurality of radiating elements in response to receiving the optical signal.
- (10) Referring to FIG. 1, a particular illustrative embodiment of an antenna is disclosed and generally designated **100**. The antenna **100** may include a plurality of radiating elements (e.g.,

radiating elements **130**, **132**, **134**, and **136**). One or more of the plurality of radiating elements **130**, **132**, **134**, and **136** may be metallic. The antenna **100** may also include a plurality of active filter nodes (e.g., active filter nodes **120**, **122**, **124**, and **126**). A particular illustrative embodiment of an active filter node is described with reference to FIG. 2.

(11) Each of the radiating elements may be coupled to a particular active filter node. For example, the radiating element **130** may be coupled to the active filter node **120**, the radiating element **132** may be coupled to the active filter node **122**, the radiating element **134** may be coupled to the active filter node **124**, and the radiating element **136** may be coupled to the active filter node **126**.

(12) The antenna **100** may include a radio frequency (RF) transmit/receive baseband section **110** coupled to an RF transmit/receive front-end section **108**. The RF transmit/receive front-end section **108** may be coupled to a feed **102** via a transmit/receive switch (or a duplex filter) **106** and via a balun **104**. The RF transmit/receive front-end section **108** may be coupled to one or more controllers (e.g., controllers **112** and **114**). In a particular embodiment, the antenna **100** may correspond to a monopole antenna.

(13) Each of the controllers **112** and **114** may be coupled to one or more light sources, such as light emitting diodes (LEDs) (e.g., LEDs **150**, **152**, **154**, and **156**). For example, the controller **112** may be coupled to the LED **150** via a current driver **140** and may be coupled to the LED **152** via a current driver **142**. As another example, the controller **114** may be coupled to the LED **154** via a current driver **144** and may be coupled to the LED **156** via a current driver **146**. In a particular embodiment, one or more of the LEDs (e.g., the LEDs **150**, **152**, **154**, and **156**) may be replaced by a laser, or another light source.

(14) In a particular embodiment, each of the light sources may be coupled, via an optical fiber, to a corresponding active filter node. For example, the LED **150** may be coupled, via an optical fiber **160**, to the active filter node **120**. The LED **152** may be coupled, via an optical fiber **162**, to the active filter node **122**. The LED **154** may be coupled, via an optical fiber **164**, to the active filter node **124**. The LED **156** may be coupled, via an optical fiber **166**, to the active filter node **126**. An optical fiber may also be referred to as a fiber optic cable. In another embodiment, the LEDs **150**, **152**, **154**, **156** may be separated from the active filter nodes **120**, **122**, **124**, **126** by free-space.

(15) In a particular embodiment, the feed **102** may be coupled to a power delivery optical laser **170** via a power delivery optical fiber **172**. In this embodiment, each of the active filter nodes **120**, **122**, **124**, and **126** may be coupled to the power delivery optical fiber **172**. The feed **102** may be powered by the power delivery optical laser **170** via the power delivery optical fiber **172**. In a particular embodiment, one or more of the active filter nodes **120**, **122**, **124**, **126** may be powered by the power delivery optical laser **170** via the power delivery optical fiber **172**, as further described with reference to FIG. 2.

(16) A distance (e.g., a wavelength separation **174**) between each of the active filter nodes **120**, **122**, **124**, or **126** and a corresponding LED **150**, **152**, **154**, or **156** may be greater than a threshold distance. The wavelength separation **174** may be sufficient such that a field of the antenna **100** is unaffected (or substantially unaffected) by the LEDs **150**, **152**, **154**, and **156** and associated drive circuits. In a particular embodiment, a frequency of operation of the antenna **100** may be based on a number of radiating elements (e.g., radiating elements **130**, **132**, **134**, and **136**) that are activated (or tuned).

(17) During operation, the controllers **112**, **114** may deactivate one or more of the radiating elements **130**, **132**, **134**, and **136** based on a particular frequency (e.g., megahertz (MHz) or gigahertz (GHz)) of operation. For example, the controller **112** may deactivate one of the radiating elements **130** and **132** based on a first frequency of operation, may deactivate both of the radiating elements **130** and **132** based on a second frequency of operation, and may deactivate neither of the radiating elements **130** and **132** based on a third frequency of operation. In a particular embodiment, the controllers **112**, **114** may select radiating elements that are farthest to the feed **102** to achieve the particular frequency of operation. The controllers **112**, **114** may receive frequency of

operation data **180** from the RF transmit/receive front-end section **108**. The frequency of operation data **180** may indicate the particular frequency of operation to which the antenna **100** is to be tuned. (18) The controllers **112**, **114** may deactivate one or more of the radiating elements **130**, **132**, **134**, and **136** using one or more of the LEDs **150**, **152**, **154**, and **156**. For example, the controllers **112**, **114** may activate a single light source or multiple light sources of the one or more light sources (e.g., the LEDs **150**, **152**, **154**, or **156**). To illustrate, the controller **112** may activate the LED **150** using the current driver **140**, may activate the LED **152** using the current driver **142**, or both. When activated, each of the LEDs **150**, **152**, **154**, and **156** may transmit an optical signal (e.g., a photonic signal **190**, **192**, **194**, or **196**) to one or more corresponding active filter nodes **120**, **122**, **124**, and **126**. In response to receiving the optical signal (e.g., the photonic signal **190**, **192**, **194**, or **196**), each of the corresponding active filter nodes **120**, **122**, **124**, and **126** may deactivate a corresponding radiating element **130**, **132**, **134**, or **136**.

(19) In a particular embodiment, each of the one or more corresponding active filter nodes **120**, **122**, **124**, and **126** may activate a corresponding radiating element **130**, **132**, **134**, or **136** in response to the optical signal (e.g., the photonic signal **190**, **192**, **194**, or **196**). In this embodiment, each of the corresponding one or more active filter nodes **120**, **122**, **124**, and **126** may deactivate the corresponding radiating element **130**, **132**, **134**, or **136** based on not receiving the optical signal (e.g., the photonic signal **190**, **192**, **194**, or **196**) for a threshold duration. For example, the active filter node **120** may receive an optical signal (e.g., the photonic signal **190**) at a first time. The active filter node **120** may activate the radiating element **130** in response to receiving the optical signal (e.g., the photonic signal **190**). Subsequently, the active filter node **120** may deactivate the radiating element **130** in response to not receiving another optical signal (e.g., the photonic signal **190**) within a threshold duration of the first time.

(20) In a particular embodiment, a particular active filter node may operate on (e.g., deactivate, activate, disconnect, connect, partially disconnect, partially connect, or change a reactance of) a single radiating element. For example, the particular active filter node (e.g., the active filter node **120**) may be coupled to two radiating elements (e.g., the radiating elements **130** and **132**). In response to receiving the photonic signal **190**, the active filter node **120** may deactivate only the radiating element **132**. In a particular embodiment, in response to receiving the photonic signal **190** the active filter node **120** may deactivate only the radiating element **130**.

(21) The optical signal (e.g., the photonic signal **190**, **192**, **194**, or **196**) may be transmitted from an LED **150**, **152**, **154**, **156** to a corresponding active filter node **120**, **122**, **124**, **126** via the corresponding optical fiber **160**, **162**, **164**, **166**. In a particular embodiment, the optical signal (e.g., the photonic signal **190**, **192**, **194**, or **196**) may be transmitted from the corresponding LED **150**, **152**, **154**, **156** to the corresponding active filter node **120**, **122**, **124**, **126** via free-space.

(22) While operating in a transmit mode of operation, the RF transmit/receive baseband section **110** may transmit a baseband signal **182** to the RF transmit/receive front-end section **108**. The RF transmit/receive front-end section **108** may generate a radio frequency (RF) signal **184** based on the baseband signal **182**. The transmit/receive switch **106** may be set to transmit in the transmit mode of operation. The RF transmit/receive front-end section **108** may transmit the RF signal **184** to the balun **104** via the transmit/receive switch **106**. The balun **104** may be used to match the antenna **100** or the one or more active filter nodes **120**, **122**, **124**, and **126**. In a particular embodiment, the antenna **100** may correspond to a dipole antenna. In a particular embodiment, the one or more active filter nodes **120**, **122**, **124**, and **126** may include a single-ended switch, a duplexer filter, or both. The balun **104** may generate a plurality of signals (e.g., RF signals **186** and **188**) based on the RF signal **184**. The balun **104** may transmit the RF signals **186** and **188** to the feed **102**.

(23) The feed **102** may transmit the RF signals **186** and **188** to one or more of the radiating elements **130**, **132**, **134**, and **136**. For example, the feed **102** may transmit the RF signal **186** to the radiating elements **130** and **132** on one side of the feed **102** and may transmit the RF signal **188** to the radiating elements **134** and **136** on the other side of the feed **102**. The activated radiating

element(s) of the radiating elements **130** and **132** may transmit the RF signal **186** over a wireless medium. For example, the activated radiating element(s) of the radiating elements **130** and **132** may radiate energy from the RF signal **186** as electromagnetic waves (radio waves) over the wireless medium. Similarly, the activated radiating element(s) of the radiating elements **134** and **136** on the other side of the feed **102** may transmit the RF signal **188** over the wireless medium. The particular frequency of operation to which the antenna **100** is tuned may be based on how many of the radiating elements **130**, **132**, **134**, and **136** are activated. For example, a higher number (e.g., 4) of activated radiating element(s) of the radiating elements **130**, **132**, **134**, and **136** may correspond to a longer antenna and a lower (e.g., 0.5 GHz) frequency of operation. As another example, a lower number (e.g., 2) of activated radiating element(s) of the radiating elements **130**, **132**, **134**, and **136** may correspond to a shorter antenna and a higher (e.g., 1 GHz) frequency of operation. The RF signals **186** and **188** may be transmitted by the activated radiating element(s) of the radiating elements **130**, **132**, **134**, and **136** at the particular frequency of operation to which the antenna **100** is tuned.

(24) In a particular embodiment, the antenna **100** may include a ground layer (e.g., a ground layer **116**) aligned with a dielectric layer (e.g., a dielectric layer **118**). For example, the dielectric layer **118** may be between the ground layer **116** and the radiating elements **130**, **132**, **134**, and **136**. The optical signal (e.g., the photonic signals **190**, **192**, **194**, or **196**) may be transmitted from the one or more light sources (e.g., the LEDs **150**, **152**, **154**, or **156**) through the dielectric layer **118** to corresponding active filter nodes (e.g., the active filter nodes **120**, **122**, **124**, or **126**). In a particular embodiment, each of the one or more light sources (e.g., the LEDs **150**, **152**, **154**, or **156**) may be located on (or embedded in) a surface (e.g., a first surface **148**) of the ground layer **116**. The first surface **148** may be adjacent to a surface (e.g., a second surface **158**) of the dielectric layer **118**. Each of the plurality of active filter nodes (e.g., the active filter nodes **120**, **122**, **124**, or **126**) may be located on another surface (e.g., a third surface **168**) of the dielectric layer **118**. The second surface **158** and the third surface **168** may be parallel to each other and may be on opposite sides of the dielectric layer **118**. To illustrate, the photonic signal **190** may be transmitted from the LED **150** through the dielectric layer **118** to the active filter node **120**. Mounting the plurality of active filter nodes (e.g., the active filter nodes **120**, **122**, **124**, or **126**) on the dielectric layer **118** may reduce the threshold distance of the wavelength separation **174**, decreasing the overall size of the antenna **100**.

(25) In a particular embodiment, the optical signal (e.g., the photonic signals **190**, **192**, **194**, or **196**) may be transmitted from the one or more light sources (e.g., the LEDs **150**, **152**, **154**, or **156**) through at least a portion of the ground layer **116** prior to being transmitted through the dielectric layer **118** to the corresponding active filter nodes (e.g., the active filter nodes **120**, **122**, **124**, or **126**). For example, the photonic signal **190** may be transmitted from the LED **150** through at least a portion of the ground layer **116** prior to being transmitted through the dielectric layer **118** to the active filter node **120**. The one or more light sources (e.g., the LEDs **150**, **152**, **154**, or **156**) may be located within the ground layer **116**, on (or embedded in) a surface of the ground layer **116**, outside the ground layer **116**, or a combination thereof.

(26) In a particular embodiment, the optical signal (e.g., the photonic signals **190**, **192**, **194**, or **196**) is transmitted via one or more light guides. For example, the photonic signal **190** may be transmitted from the LED **150** through the dielectric layer **118** via a light guide to the active filter node **120**. As another example, the photonic signal **190** may travel from the LED **150** through at least a portion of the ground layer **116** via a light guide prior to being transmitted through the dielectric layer **118** to the active filter node **120**.

(27) In a particular embodiment, the optical signal (e.g., the photonic signals **190**, **192**, **194**, or **196**) travels in a substantially straight path from the one or more light sources (e.g., the LEDs **150**, **152**, **154**, or **156**) to the corresponding active filter nodes (e.g., the active filter nodes **120**, **122**, **124**, or **126**). In an alternative embodiment, the optical signal (e.g., the photonic signals **190**, **192**, **194**, or **196**) travels in a curved (or angular) path from the one or more light sources (e.g., the LEDs **150**,

152, 154, or 156) to the corresponding active filter nodes (e.g., the active filter nodes **120, 122, 124, or 126**). For example, the photonic signal **190** may travel via a curved light guide through at least a portion of the ground layer **116**, the dielectric layer **118**, or both.

(28) In a particular embodiment, the ground layer **116** includes at least a portion of one or more components of the antenna **100**. For example, the ground layer **116** may include at least a portion of one or more of the current drivers **140, 142, 144, 146**, the controllers **112, 114**, the transmit/receive switch **106**, the RF transmit/receive front-end section **108**, the transmit/receive baseband section **110**, the balun **104**, the one or more light sources (e.g., the LEDs **150, 152, 154, or 156**), the feed **102**, the plurality of active filter nodes (e.g., the active filter nodes **120, 122, 124, or 126**), or the power delivery optical laser **170**.

(29) In a particular embodiment, the dielectric layer **118** may include at least a portion of one or more components of the antenna **100**. For example, the dielectric layer **118** may include at least a portion of one or more of the current drivers **140, 142, 144, 146**, the controllers **112, 114**, the transmit/receive switch **106**, the RF transmit/receive front-end section **108**, the transmit/receive baseband section **110**, the balun **104**, the one or more light sources (e.g., the LEDs **150, 152, 154, or 156**), the feed **102**, the plurality of active filter nodes (e.g., the active filter nodes **120, 122, 124, or 126**), or the power delivery optical laser **170**.

(30) The controllers **112, 114** may deactivate one or more of the radiating elements **130, 132, 134, 136** using optical signals based on a particular frequency of operation of the antenna **100**. Having the dielectric layer **118** between the one or more light sources (e.g., the LEDs **150, 152, 154, or 156**) and the corresponding active filter nodes (e.g., the active filter nodes **120, 122, 124, or 126**) may reduce the overall size of the antenna **100**, making the antenna **100** suitable for miniaturized manufacturing techniques.

(31) Referring to FIG. 2, an active filter node is disclosed and generally designated **200**. In a particular embodiment, the active filter node **200** corresponds to one of the active filter nodes **120, 122, 124, or 126** of FIG. 1. The active filter node **200** may include a capacitively-coupled radio frequency (RF) bi-directional active filter node. The active filter node **200** includes an amplifier **220** coupled to a radiating element **230** on one side of the amplifier **220** and to a radiating element **232** on another side of the amplifier **220**. In a particular embodiment, each of the radiating elements **230** and **232** corresponds to one of the radiating elements **130, 132, 134, and 136** of FIG. 1. In a particular embodiment, the amplifier **220** may include a capacitively-coupled amplifier, a bi-directional amplifier, a radio frequency amplifier, or any combination thereof. The active filter node **200** may, for example, include a Bessel filter transfer function. The active filter node **200** may implement a bi-quadratic transfer function which may synthesize other filters. The active filter node **200** may include a splitter, a photo-voltaic detector, a rectifying photo-voltaic detector, or a combination thereof (e.g., a splitter and photo-voltaic detector **224**). The active filter node **200** may be coupled to the power delivery optical fiber **172** via the splitter and photo-voltaic detector **224**. In a particular embodiment, the splitter and photo-voltaic detector **224** may include a splitter and a rectifying photo-voltaic detector. The active filter node **200** may include a switch **222**. In a particular embodiment, the switch **222** may include a bi-polar transistor, a field effect transistor, an analog photo-responsive switch, an analog photo-responsive resistor, or a combination thereof. The switch **222** may be coupled to an optical fiber **260**. In a particular embodiment, the optical fiber **260** corresponds to one of the optical fibers **160, 162, 164, or 166** of FIG. 1.

(32) During operation, the amplifier **220** may receive a photonic signal notification **202** from the switch **222**. For example, the switch **222** may transmit the photonic signal notification **202** to the amplifier **220** in response to receiving a photonic signal (e.g., the photonic signal **190, 192, 194, or 196** of FIG. 1). In a particular embodiment, the active filter node **200** corresponds to the active filter node **120** of FIG. 1. In this embodiment, the switch **222** transmits the photonic signal notification **202** to the amplifier **220** in response to receiving the photonic signal **190** of FIG. 1. The amplifier **220** deactivates one or more of the radiating element **230, 232** in response to receiving

the photonic signal notification **202**. In another particular embodiment, the amplifier **220** may activate one or more of the radiating element **230**, **232**, in response to receiving the photonic signal notification **202**.

(33) In a particular embodiment, the amplifier **220** is powered by the splitter and photo-voltaic detector **224**. For example, the splitter and photo-voltaic detector **224** may generate a current **204** based on detecting transmission of light through the power delivery optical fiber **172** and may transmit the current **204** to the amplifier **220**.

(34) Thus, the antenna **100** may include the active filter node **200** with the amplifier **220**. Use of the active filter node **200** may replace use of trap in the antenna **100**. The active filter node **200** may enable the antenna **100** to operate at a higher frequency bandwidth than the trap.

(35) FIG. **3** is a flowchart to illustrate a particular embodiment of a method **300** of optically tuning an antenna. In an illustrative embodiment, the method **300** may be performed to tune the antenna **100** of FIG. **1**.

(36) The method **300** includes determining a configuration of an antenna based on a particular frequency, at **302**. For example, in FIG. **1**, the controllers **112**, **114** may determine a configuration of the antenna **100** based on a particular frequency using frequency and configuration correlation data. To illustrate, the frequency and configuration correlation data may map specific frequencies (e.g., 1 GHz) to a particular number (e.g., 2) of deactivated (or activated) radiating elements of the plurality of radiating elements (e.g., the radiating elements **130**, **132**, **134**, and **136**. Each of the radiating elements **130**, **132**, **134**, and **136** may have an equivalent effect on the frequency of operation of the antenna **100**.

(37) For example, a frequency range (e.g., spanning 500 MHz to 2 GHz) may correspond to a particular dipole overall antenna length (e.g., approximately 12 inches). Each of the radiating elements **130**, **132**, **134**, and **136** may correspond to a particular antenna length portion (e.g., 3 inches) when activated. A number (e.g., $12/3=4$) of activated radiating elements corresponding to a specific frequency (e.g., 500 MHz) may be based on the particular overall antenna length (e.g., approximately 12 inches) and the particular antenna length portion (e.g., 3 inches). The number (e.g., $4-4=0$) of deactivated radiating elements corresponding to the specific frequency (e.g., 500 MHz) may be based on the number (e.g., 4) of activated radiating elements and a total number (e.g., 4) of radiating elements of the antenna **100**.

(38) In a particular embodiment, any of the radiating elements **130**, **132**, **134**, and **136** adding up to the particular number are selectable by the controllers **112**, **114** to achieve the particular frequency. In an alternative embodiment, the particular number of radiating elements that are farthest (or nearest) to the feed **102** are selectable by the controllers **112**, **114** to achieve the particular frequency.

(39) In a particular embodiment, the frequency and configuration correlation data maps the specific frequencies to particular deactivated radiating elements (e.g., the radiating elements **132** and **136**), to particular activated radiating elements (e.g., the radiating elements **130** and **134**), or to both. A particular radiating element (e.g., the radiating element **132**) may affect the frequency of operation of the antenna **100** differently than another radiating element (e.g., the radiating element **130**) when activated. In a particular embodiment, the controllers **112**, **114** may receive the frequency and configuration correlation data from the RF transmit/receive front-end section **108**.

(40) The method **300** also includes selecting a first light source of a plurality of light sources based on the configuration, at **304**. For example, in FIG. **1**, the controller **112** may select the LED **152** and the controller **114** may select the LED **156** based on the particular number (e.g., 2) of deactivated radiating elements indicated by the configuration.

(41) The method **300** further includes activating the first light source. The first light source, in response to being activated, may transmit an optical signal to a first active filter node of a plurality of active filter nodes of the antenna. Each of a plurality of radiating elements of the antenna may be coupled to a particular active filter node of the plurality of active filter nodes. The first active filter

node may deactivate a corresponding radiating element of the plurality of radiating elements in response to the optical signal. For example, in FIG. 1, the controller **112** may activate the LED **152** and the controller **114** may activate the LED **156**. The LED **152** may transmit the photonic signal **192** to the active filter node **122** and the LED **156** may transmit the photonic signal **196** to the active filter node **126**. The active filter node **122** may deactivate the radiating element **132** in response to receiving the photonic signal **192** and the active filter node **126** may deactivate the radiating element **136** in response to receiving the photonic signal **196**.

(42) Thus, the method **300** may enable optical tuning of the antenna **100** to the particular frequency.

(43) FIG. 4 is a block diagram illustrates an embodiment of a general computer system that is generally designated **400**. The computer system **400** may be operable to support embodiments of computer-implemented methods, computer program products, and system components as illustrated in FIGS. 1-3. The computer system **400**, or any portion thereof, may operate as a standalone device or may be activated, e.g., using a network, to other computer systems or peripheral devices.

(44) The computer system **400** may be incorporated into a mobile computing device. The mobile computing device may include the antenna **100** of FIG. 1. The antenna **100** may be controlled by a controller **428** of the computer system **400**.

(45) The computer system **400** can also be implemented as or incorporated into various devices, such as a tablet PC, a personal digital assistant (PDA), a palmtop computer, a laptop computer, a communications device, a web appliance, a display device, a computing device, or any other machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while a single computer system **400** is illustrated, the term “system” shall also be taken to include any collection of systems or sub-systems that individually or jointly execute a set, or multiple sets, of instructions to perform one or more computer functions.

(46) As illustrated in FIG. 4, the computer system **400** may include a processor **402**, e.g., a central processing unit (CPU). In a particular embodiment, the processor **402** may include multiple processors. For example, the processor **402** may include distributed processors, parallel processors, or both. The multiple processors may be included in, or coupled to, a single device or multiple devices. The processor **402** may be used to support a virtual processing environment. In a particular embodiment, the processor **402** may include a state machine, an application specific integrated circuit (ASIC), or a programmable gate array (PGA) (e.g., a field programmable gate array (FPGA)).

(47) Moreover, the computer system **400** may include a main memory **404** and a static memory **406** that may communicate with each other via a bus **408**. In a particular embodiment, the main memory **404** may include processor-executable instructions **424**. As shown, the computer system **400** may further include or be coupled to a display unit **410**, such as a liquid crystal display (LCD), an organic light emitting diode (OLED), a flat panel display, a solid-state display, or a projection display. Additionally, the computer system **400** may include an input device **412**, such as a keyboard, a remote control device, and a cursor control device **414**, such as a mouse. In a particular embodiment, the cursor control device **414** may be incorporated into the remote control device. The computer system **400** may also include a disk drive unit **416**, a signal generation device **418**, such as a speaker, and a network interface device **420**. The network interface device **420** may be coupled to other devices (not shown) via a network **426**.

(48) In a particular embodiment, one or more of the components of the computer system **400** may be coupled to an antenna or an antenna system (e.g. the antenna **100** or a system coupled to the antenna **100** of FIG. 1). In a particular embodiment, the controller **428** may correspond to the controller **112** of FIG. 1, the controller **114** of FIG. 1, or both.

(49) In a particular embodiment, as depicted in FIG. 4, the disk drive unit **416** may include a tangible computer-readable storage device **422** in which one or more sets of instructions **424**, e.g. software, may be embedded. Further, the instructions **424** may embody one or more of the methods

or logic as described herein. The processor **402** may execute the instructions **424** to perform operations corresponding to one or more of the methods or logic as described herein. The processor **402** may perform the operations directly, or the processor **402** may facilitate, direct, or cooperate with another device or component to perform the operations.

(50) In a particular embodiment, the instructions **424** may reside completely, or at least partially, within the main memory **404**, the static memory **406**, and/or within the processor **402** during execution by the computer system **400**. The main memory **404** and the processor **402** also may include tangible computer-readable media.

(51) In an alternative embodiment, dedicated hardware implementations, such as application specific integrated circuits, programmable logic arrays and other hardware devices, can be constructed to implement one or more of the methods described herein. Applications that may include the apparatus and systems of various embodiments can broadly include a variety of electronic and computer systems. One or more embodiments described herein may implement functions using two or more specific interconnected hardware modules or devices with related control, or as portions of an application-specific integrated circuit. Accordingly, the present system encompasses software, firmware, and hardware implementations.

(52) In accordance with various embodiments of the present disclosure, the methods described herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limiting embodiment, implementations can include distributed processing and parallel processing. Alternatively, virtual computer system processing can be used to implement one or more of the methods or functionality as described herein.

(53) The present disclosure describes a computer-readable non-transitory medium that includes instructions **424** so that an antenna may be optically tuned. Further, the instructions **424** may be transmitted or received over the network **426** via the network interface device **420** (e.g., via uploading and/or downloading of an optical antenna tuning application or program, or both).

(54) While the computer-readable non-transitory medium is shown to be a single medium, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “non-transitory computer-readable medium” shall also include any medium that is capable of storing a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

(55) In a particular non-limiting, exemplary embodiment, the computer-readable non-transitory medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable non-transitory medium can be a random access memory or other volatile re-writable memory. Additionally, the computer-readable non-transitory medium can include a magneto-optical or optical medium, such as a disk or tapes. Accordingly, the disclosure is considered to include any one or more of a computer-readable non-transitory storage medium and successor media, in which data or instructions may be stored.

(56) It should also be noted that software that implements the disclosed methods may optionally be stored on a tangible storage medium, such as: a magnetic medium, such as a disk or tape; a magneto-optical or optical medium, such as a disk; or a solid state medium, such as a memory card or other package that houses one or more read-only (non-volatile) memories, random access memories, or other re-writable (volatile) memories.

(57) Although the present specification describes components and functions that may be implemented in particular embodiments with reference to particular standards and protocols, the claims are not limited to such standards and protocols. For example, standards for Internet, other packet switched network transmission and standards for viewing media content represent examples of the state of the art. Such standards are periodically superseded by faster or more efficient equivalents having essentially the same functions. Accordingly, replacement standards and protocols having the same or similar functions as those disclosed herein are considered equivalents

thereof.

(58) Moreover, although specific embodiments have been illustrated and described herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the description.

(59) The Abstract of the Disclosure is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure. This disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. As the following claims reflect, inventive subject matter may be directed to less than all of the features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

(60) The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments, which fall within the scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

Claims

1. An antenna system comprising: a dielectric layer; a first filter node coupled to a first surface of the dielectric layer; a second filter node coupled to the first surface of the dielectric layer; a first radiating element coupled to the first filter node and to the second filter node; a second radiating element coupled to the second filter node; an optical fiber coupled to the first filter node and to the second filter node, the optical fiber configured to carry a first light signal to power the first filter node and the second filter node; a first light source embedded in a ground layer located adjacent to a second surface of the dielectric layer, the first light source configured to transmit a second light signal to the second filter node to cause the second filter node to adjust a characteristic of the second radiating element using a portion of the power supplied by the first light signal, wherein a thickness of the dielectric layer contributes to a wavelength separation between the first light source and the second filter node, the wavelength separation being greater than a threshold distance to reduce any effect of the first light source on a field of the antenna; and a controller coupled to the first light source, the controller configured to activate the first light source to enable, or disable, use of the second radiating element as part of an antenna at a particular operating frequency based on the second light signal.
2. The antenna system of claim 1, wherein the second light signal produced by the first light source causes the second filter node to enable use of the second radiating element coupled to the second filter node.
3. The antenna system of claim 1, wherein the second light signal produced by the first light source causes the second filter node to disable use of the second radiating element coupled to the second filter node.
4. The antenna system of claim 1, further comprising a second light source configured to transmit a third light signal to the first filter node to enable use of the first radiating element and the second radiating element.
5. The antenna system of claim 1, further comprising a second light source configured to transmit a

third light signal to the first filter node to disable use of the first radiating element and the second radiating element.

6. The antenna system of claim 1, wherein the second filter node comprises an inductance, a capacitance, a gain, a reactance, or a combination thereof.

7. The antenna system of claim 1, wherein the second filter node comprises an amplifier.

8. The antenna system of claim 7, wherein the second light signal is received by a switch of the second filter node, and wherein the switch is coupled to control the amplifier.

9. The antenna system of claim 8, wherein the amplifier of the second filter node uses the power from the first light signal for enabling or disabling the second radiating element.

10. An antenna system comprising: a dielectric layer; a first filter node coupled to a first surface of the dielectric layer; a second filter node coupled to the first surface of the dielectric layer; a first radiating element coupled to the first filter node and to the second filter node; a second radiating element coupled to the second filter node; an optical fiber coupled to the first filter node and to the second filter node, the optical fiber configured to carry a power light signal to power the first filter node and the second filter node; a first light source configured to transmit a first light signal to the first filter node to cause the first filter node to enable or disable both the first radiating element and the second radiating element; a second light source configured to transmit a second light signal to the second filter node to cause the second filter node to enable or disable the second radiating element; a processing system including a processor coupled to the first light source and the second light source; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: determining a configuration of the antenna system based on an operating frequency; and activating at least one of the first light source and the second light source based to place the antenna system into the configuration.

11. The antenna system of claim 10, wherein the second filter node comprises an inductance, a capacitance, a gain, a reactance, or a combination thereof.

12. The antenna system of claim 10, wherein the second filter node comprises an amplifier.

13. The antenna system of claim 12, wherein the second light signal is received by a switch of the second filter node, and wherein the switch is coupled to control the amplifier.

14. The antenna system of claim 12, wherein the amplifier of the second filter node uses the power from the power light signal for enabling or disabling the second radiating element.

15. An antenna system comprising: a dielectric layer; a plurality of filter nodes coupled to a first surface of the dielectric layer; a plurality of radiating elements coupled between pairs of filter nodes of the plurality of filter nodes; an optical fiber coupled to the plurality of filter nodes, the optical fiber configured to carry a power light signal to power the plurality of filter nodes; a plurality of light sources configured to transmit respective light signals to respective ones of the plurality of filter nodes to cause the respective ones of the plurality of filter nodes to enable or disable respective ones of the plurality of radiating elements; a processing system including a processor coupled to the plurality of light sources; and a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising: determining a configuration of the antenna system based on an operating frequency; and activating at least one of the plurality of light sources to place the antenna system into the configuration.

16. The antenna system of claim 15, wherein each filter node of the plurality of filter nodes comprises an inductance, a capacitance, a gain, a reactance, or a combination thereof.

17. The antenna system of claim 15, wherein each filter node of the plurality of filter nodes comprises an amplifier.

18. The antenna system of claim 17, wherein each filter node of the plurality of filter nodes comprises a switch, and wherein the respective light signals are received by the switch, and wherein the switch is coupled to control the amplifier.

19. The antenna system of claim 17, wherein the amplifier uses the power from the power light signal for enabling or disabling the respective ones of the plurality of radiating elements.

20. The antenna system of claim 17, wherein the activating the at least one of the plurality of light sources comprises disabling at least one of the plurality of radiating elements to shorten a length of the antenna system.
